

Super Node Analysis

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SENSE

Super node

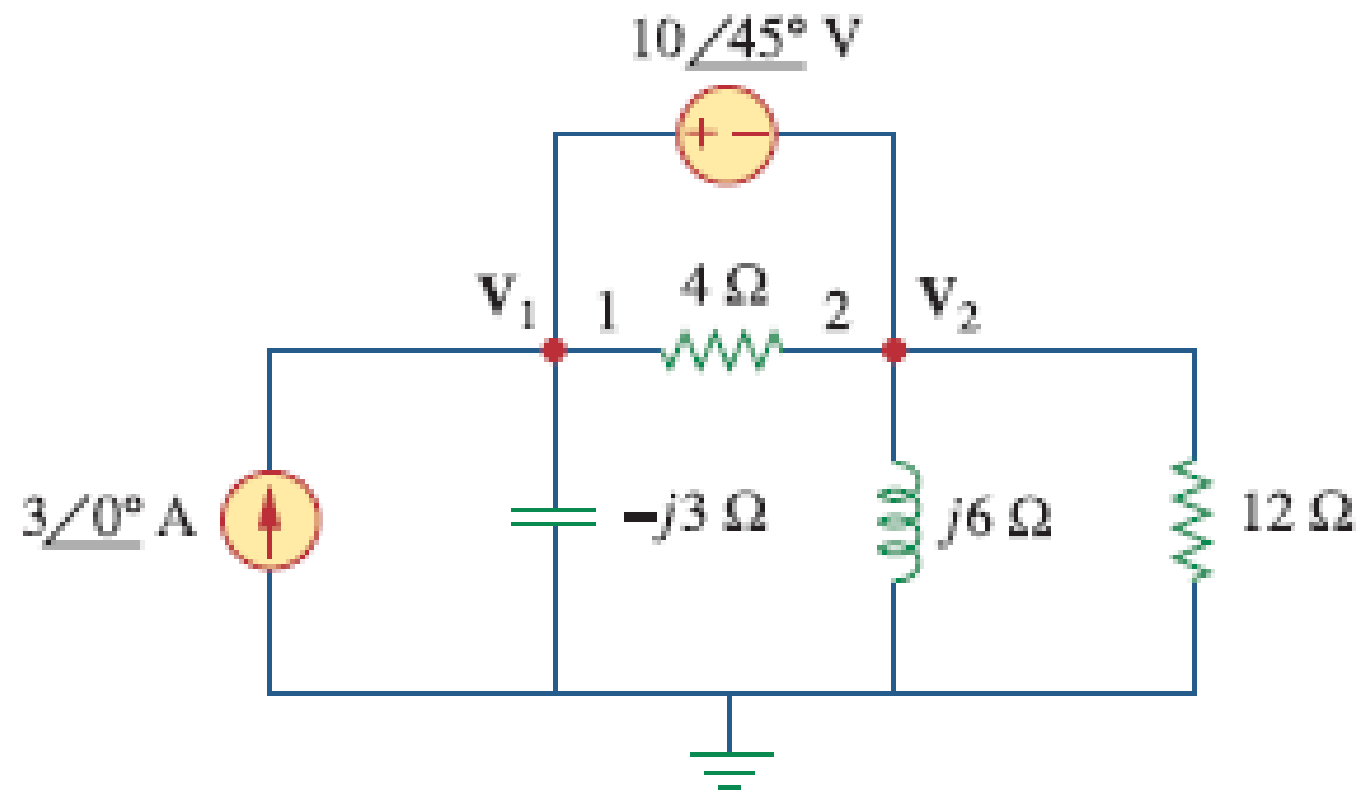
A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.

Apply KCL to the supernode by treating both nodes a single supernode while removing any element in parallel to it.

Apply KVL to the supernode while removing all elements connected to it.

Super node

Compute V_1 and V_2 in the circuit



Solution:

Nodes 1 and 2 form a supernode

Applying KCL at the supernode gives

$$3 = \frac{\mathbf{V}_1}{-j3} + \frac{\mathbf{V}_2}{j6} + \frac{\mathbf{V}_2}{12}$$

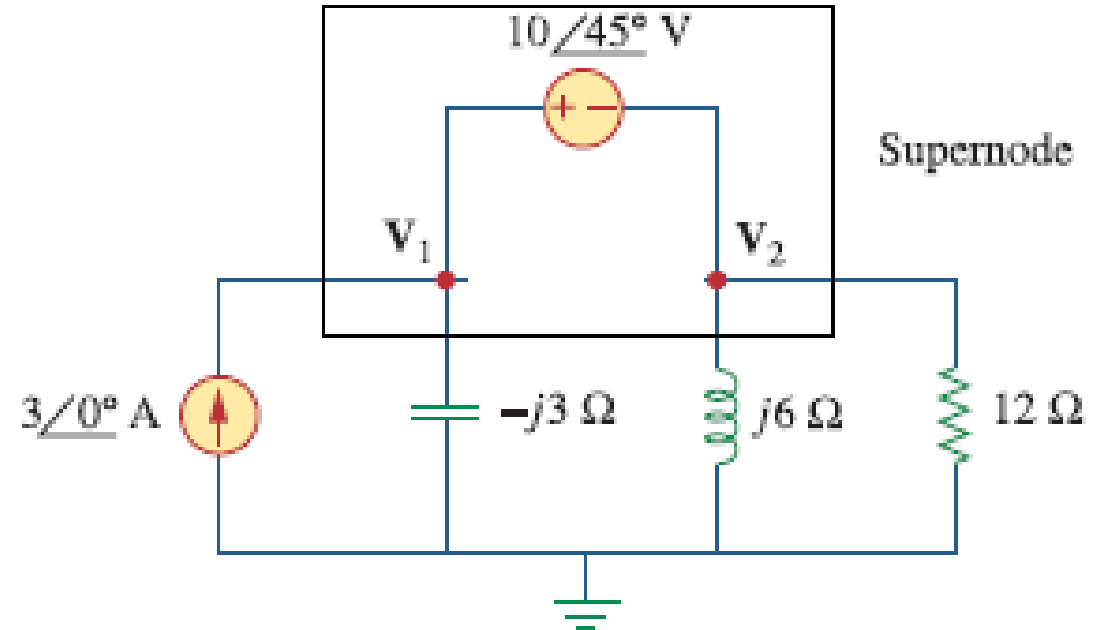
$$36 = j4\mathbf{V}_1 + (1 - j2)\mathbf{V}_2$$

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ$$

$$36 - 40\angle 135^\circ = (1 + j2)\mathbf{V}_2$$

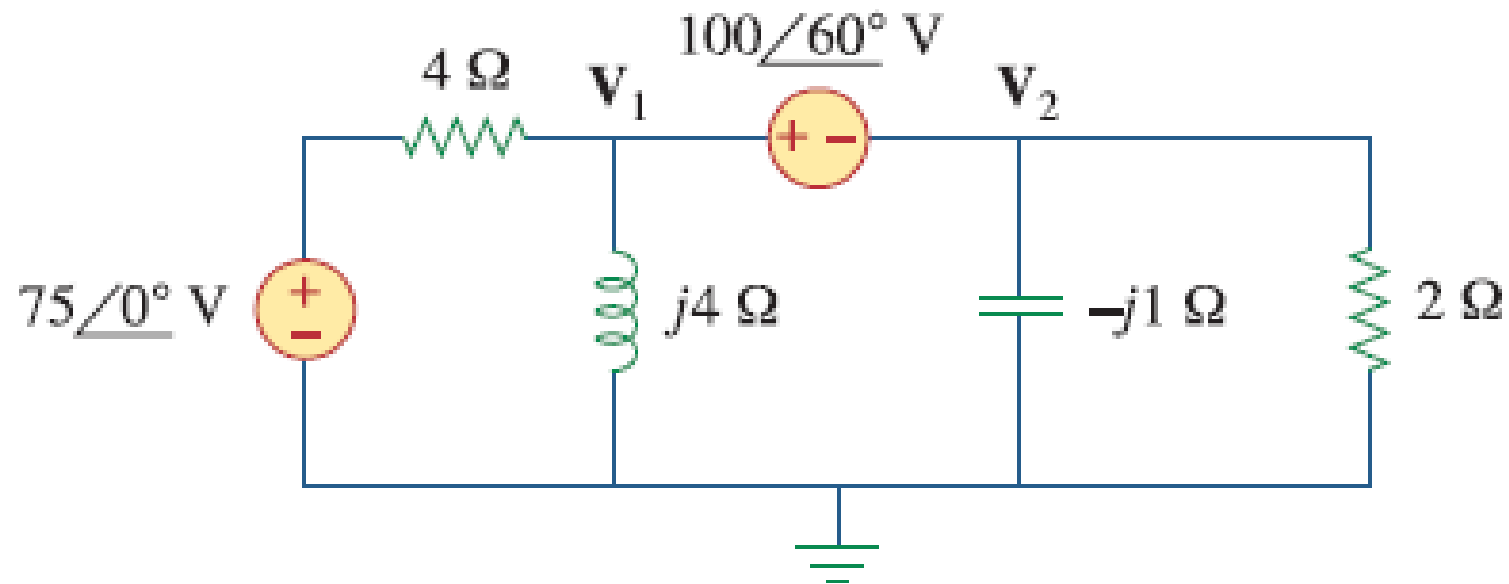
$$\mathbf{V}_2 = 31.41\angle -87.18^\circ \text{ V}$$

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$



- Home work Problem -1:

Calculate V_1 and V_2 in the circuit



Answer: $V_1 = 96.8\angle 69.66^\circ \text{ V}$, $V_2 = 16.88\angle 165.72^\circ \text{ V}$.